



International Islamic University Chittagong

Faculty of Shari'ah & Islamic Studies

Part A: Fundamental Information about this Course

1	CourseCode:	ISCED Code
	STAT-3503	0542
2	Course Title:	Statistics for Research in Shari'ah Studies
3	Course Type:	Mathematical Economics
4	Semester:	Sixth
5	Prerequisites Course:	Money and Banking in Islamic Perspectives
6	Contact Hours:	3(Three)
7	Credit Hours:	3(Three)
8	Total Marks:	100
9	Rationale of the course:	

Course Assessments	CIE: Continuous Internal Evaluation	Attendance	10 Marks
		Class test/ Assignment/ Quizzes	10 Marks
		Midterm	30 Marks
	SFE: Semester Final Examination		50 Marks

Part B: Course Objectives

The course is designed to equip students with conceptual introduction to the field of econometrics and many of its interpretations and applications, which is a must especially for the students of economics. This course provides knowledge on primary statistical, mathematical and economic measurement, analysis and their interpretation and application in business and practical life. In this regard, the course will provide knowledge on data, frequency distribution, the central tendency, variability of data, skewness, moments, kurtosis of any distribution and index number.

Part C: Course Learning Outcomes and Mapping of Course Learning Outcomes (CLO) to Program Learning Outcomes (PLO):

This course is designed to equip students with knowledge of different statistical tools used in business analysis. This course provides knowledge on primary statistical, mathematical and economic measurement, analysis and their application in business. This course will provide knowledge on types of data, organize the collected data into tabular format and provide a descriptive and inferential analysis using learnt tools. This course will serve as the foundation of statistics and data analysis for research in future. **At the end of the Course, the Student will be able to-**

S/N	Course Learning Outcomes (CLOs): Upon the successful completion of the course, students will be able to	Corresponding PLO	Bloom's taxonomy domain/level
CLO1	Demonstrate knowledge of fundamental statistical, mathematical and economic terms, concepts, and skills needed to describe data sets using graphs and numerical summaries; distinguish between theoretical and empirical.	PLO-1	Cognitive/ Understanding and analyzing
CLO2	apply mathematical concepts and principles to perform numerical and symbolic computations.	PLO-3	Cognitive/ Applying
CLO3	Solve mathematical and statistical problems, use mathematics and statistics independently, and develop the ability to analyze, interpret, and apply quantitative information.	PLO-4	Cognitive/ Applying

C. Summary of Course Contents

SI No.	COURSE CONTENT (as Summary)	Hrs	CLOs
	Section-A (Mid-term: 30 Marks)		
1	Chapter One: Introduction to Statistics: Meaning of Statistics, Types of Statistics, Functions of Statistics, and Characteristics of Statistics. Sample and Population, Importance of Statistics in Economics, Limitations of Statistics. Chapter Two: Data Collection Methods: Statistical Data, Primary and Secondary Data, Methods of collecting primary data: Observation, Survey, Interview, Through Schedule, through questionnaire, Characteristics of a good questionnaire, Guidelines for constructing a questionnaire, Online/Internet (E-mail/Web) and Case study method, Secondary data - Meaning of secondary data, Sources of secondary data, Comparison between primary and secondary data,	9 Lecture, White Board Writing, PowerPoint Presentation, Q/A, Class discussion	CLO1
2	Chapter Three: Data Classification and Presentation: Classification of Data, Tabulation of Data. Frequency Distribution, Steps of Making Frequency Distribution, Cumulative Frequency Distribution, Diagrams and Graphs of Statistical Data. Chapter Four: Measures of Central Tendency Average and Types of Average, Objectives of Averaging, Characteristics of a Good Average, Arithmetic Mean, Properties of Arithmetic Mean.,	9 Lecture, White Board Writing, PowerPoint Presentation, Q/A, Class discussion	CLO1

	Concept of Mode, Advantage and Disadvantage of Mode, Concept of Median.		
	Section-B (SEE: 50 Marks)		
4	Chapter Five: Measures of Dispersion/Variation Introduction to Measures of Dispersion, Absolute Measures of Dispersion, Relative Measures of Dispersion, Range and Co-efficient of Range, Mean-deviation and its Co-efficient, Standard Deviation, Variance, Co-efficient of Standard Deviation and Variation Chapter Six: Calculus and its Applications: First-Order Differential Equation with Constant Coefficient and Constant Term, First-Order Differential Equation with Variable Coefficient and Variable term, economic applications and examples: linear model in economics, market model, national income analysis, Application of Differential Equations in Economics: Dynamics of Market Price, cost, revenue, profit calculation techniques, mathematical methods and techniques formulated in abstract settings to concrete economic applications.	9	CLO2
5	Chapter Seven: Sampling & Sampling Distribution: Population, Sample, Parameter, Statistic, Sampling theory, draw a simple random sample by Random Number method, calculate its mean and standard error, and determine sample size. Sampling distribution of 'Z', 't', statistic, sample mean and proportion and its applications in Economics and Banking,	6	CLO-3
6	Chapter Eight: Introduction to Econometrics: Definition of econometrics, economic, and econometric models, nature and methodology of econometrics, Functions of econometrics, Characteristics of econometrics, Importance of econometrics in Shariah research. The Linear Regression Analysis and Hypothesis: An Econometric Model; Introducing the error term; Estimating the Regression Parameters: The Least Squares Principle, Interpreting the Estimates; Other Economic Models, Multiple Regression Analysis,	6 Lecture, White Board Writing, Powerpoint Presentation, Q/A, Class discussion	CLO-4
	Chapter Nine: Research Project: Selection of a research problem/title, Formulation the research hypothesis, Research design, Questionnaire design, Sampling design, Survey design, Data collection (primary, secondary), Processing the data, Analysis of	6	CLO-3

	data (Descriptive and Inferential or Statistical analysis) using statistical software such as: MS Excel, SPSS, etc. and Research report writing and Presentation of the report.	Field work and lab work	
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D. Learning Resources

▪ **Textbook/ Reference Materials**

- Gupta, S. P., & Gupta, M. P. (2010), Business statistics. Sultan Chand & Sons, India
- Anderson, D. R., Sweeney, D. J., & Williams, T. A. (2011) Statistics for Business and Economics, 11th Edition, South-Western Cengage Learning
- Kothari, C. R. (2004). Research methodology: Methods and techniques. Ed. 2nd, New Age International, Jaipur, India
- Gujarati, M., Damodar (2012), Basic Econometrics, Sultan Chand & Sons, India
- Chiang, Alpha C (2001), Fundamentals of Mathematical Economics, MC Grow Hill, UK

E. Course Assessment Pattern (Theory courses):

Bloom's Category		Evaluations out of 100 marks			
		CIE (50 marks)			SEE (50marks)
Cognitive learning	Affective Learning	Mid-term: (30)	Assignment/ Class Test: (10)	Attendance Marks (:10)	Written Exam: (50)
Remember	-	-	-	-	5
Understand	-	5	5	-	5
Apply	-	5	-	-	10
Analyze	-	5	-	-	10
Evaluation	-	10	5	-	10
Create	-	5	-	-	10
x	Responding	x	x	10	
Remarks	Course teachers may change the magnitude of marks in Bloom's category (Both for CIE and SEE), but he/she will have to keep in mind that the % of higher order learning mode must be about 60% or more and all the Bloom's categories to be addressed during the semester.				

- **Note:** CIE=Continuous Internal Evaluation, SEE= Semester End Examination.
- **Delivery methods & activities:** Lecture, White Board Writing, Questions and Answers, Discussions Power point Presentation,
- **Assessment tools:** Class Attendance, Class test, Quizzes/ Assignment. Mid-Term & Final Exam. Project evaluation & Viva.

